



TEACHABLE MACHINE

Group Exercise

TEACHABLE MACHINE

<https://teachablemachine.withgoogle.com/>



OBJECTIVE

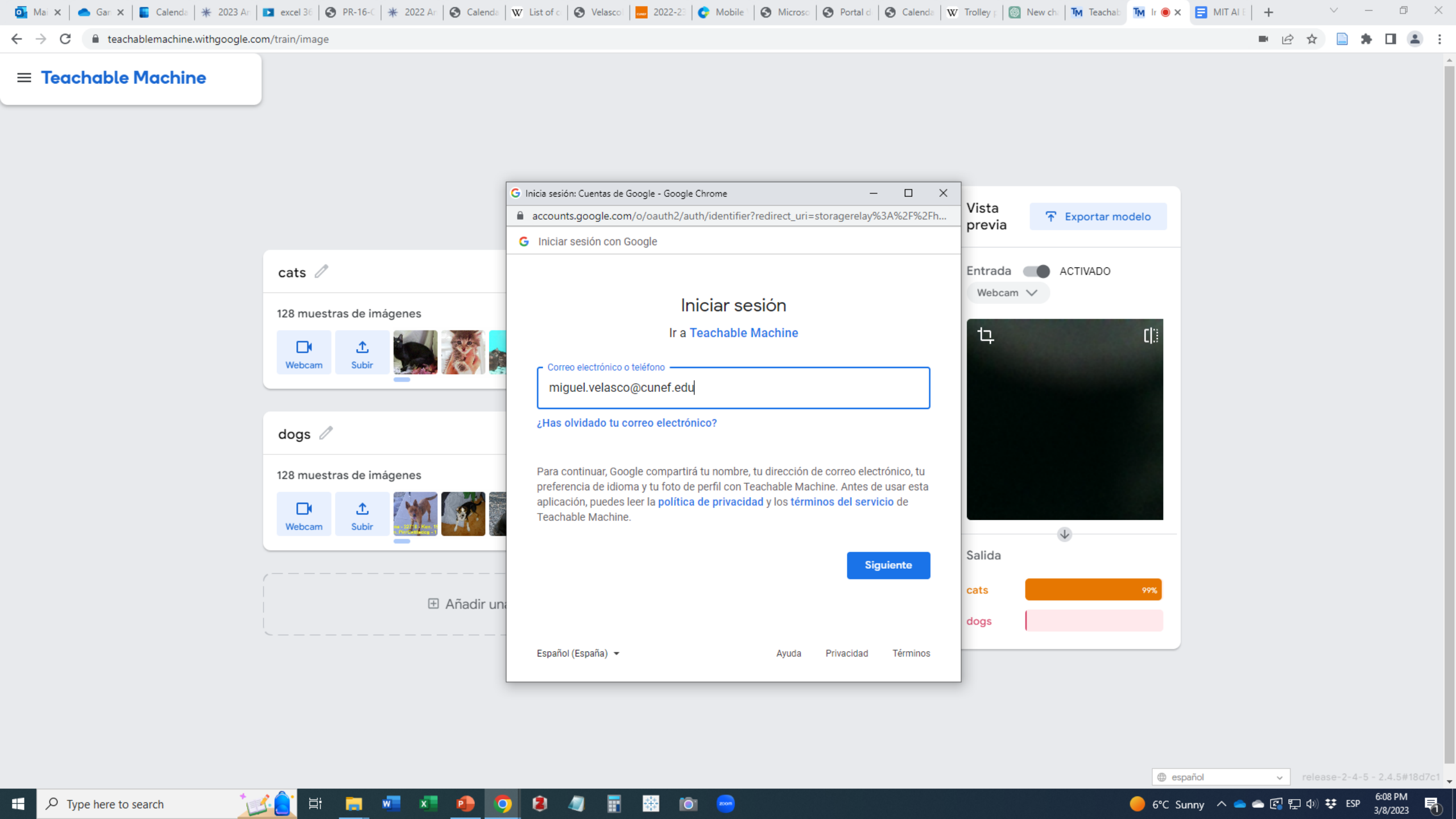
- Train your own neural network with something that has a real-world application (however modest).
- You must choose a type of project
 1. Image recognition
 2. Sound recognition
 3. Pose recognition

OBJECTIVE

- You must (successfully) train a neural network and present your work to the rest of the class.
- This is a group activity (you all have a group assigned to the group project, so we will use those same groups)
- One person from each group must upload the project through Canvas

METHODOLOGY

- You have to choose a real application in which you will use teachable machine.
- Then you will have to train the algorithm and test it with inputs that I have not seen until now (as I have done in class) and improve it if necessary.
- On the day we determine, we will dedicate the class to groups presenting their models and explaining how they can fit into an application in the real world, including a demonstration of how they work live (remember to upload the link of the model to Canvas before class!).



cats

128 muestras de imágenes



Webcam



Subir



dogs

128 muestras de imágenes



Webcam



Subir



Añadir una

Inicia sesión: Cuentas de Google - Google Chrome

accounts.google.com/o/oauth2/auth/identifier?redirect_uri=storagerelay%3A%2F%2Fh...

Iniciar sesión con Google

Iniciar sesión

Ir a Teachable Machine

Correo electrónico o teléfono

[¿Has olvidado tu correo electrónico?](#)

Para continuar, Google compartirá tu nombre, tu dirección de correo electrónico, tu preferencia de idioma y tu foto de perfil con Teachable Machine. Antes de usar esta aplicación, puedes leer la [política de privacidad](#) y los [términos del servicio](#) de Teachable Machine.

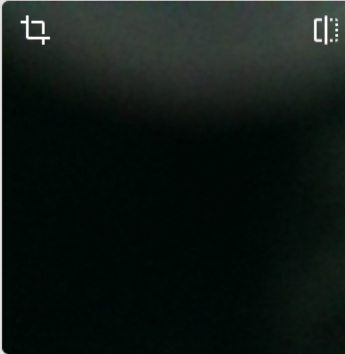
Siguiente

Español (España) Ayuda Privacidad Términos

Vista previa **Exportar modelo**

Entrada ☒ ACTIVADO

Webcam



Salida

cats 99%

dogs

Teachable Machine

+ Nuevo proyecto

gatoperros  

 Abrir proyecto desde Drive

 Guardar proyecto en Drive

 Ver proyecto en Drive

 Hacer una copia en Drive

 Cerrar sesión de Drive

 Abrir el proyecto desde un archivo

 Descargar proyecto como archivo

 Información de Teachable Machine

 Preguntas frecuentes

1. Recoger muestras

2. Preparar el modelo

3. Exportar el modelo

 Enviar comentarios

cats 

128 muestras de imágenes

Webcam

Subir

















dogs 

128 muestras de imágenes

Webcam

Subir

















Añadir una clase

Preparación

Preparar modelo

Avanzado

Vista previa

Exportar modelo

Para obtener una vista previa de un modelo aquí, primero debes prepararlo en la parte de la izquierda.

Teachable Machine

Ruido de fondo ?

20 muestras de audio / 20 como mínimo

Micrófono

Subir

hola

15 muestras de audio / 8 como mínimo

Micrófono

Subir

adios

15 muestras de audio / 8 como mínimo

Micrófono

Subir

Añadir una clase

Preparación

Modelo preparado

Avanzado

Vista previa

Export Model

Input

OFF

Overlap Factor: 0.75

Output

Ruido de fondo

hola

adios

To export

English

release-2-4-4 - 2.4.4#95c54c

Click here to generate the link to the file containing the project

Export your model to use it in projects.

Tensorflow.js ⓘ

Tensorflow Lite ⓘ

Export your model:

☒ Upload (shareable link)

☐ Download

Your sharable link:

[https://teachablemachine.withgoogle.com/models/\[...\]](https://teachablemachine.withgoogle.com/models/[...])

When you upload your model, Teachable Machine hosts it at this link for free. (FAQ: [Who can use my model?](#))

Code snippets to use your model:

JavaScript

p5.js

[Contribute on Github](#)

Learn more about how to use the code snippet on [github](#).

```
<div>Teachable Machine Audio Model</div>
<button type="button" onclick="init()">Start</button>
<div id="label-container"></div>
<script src="https://cdn.jsdelivr.net/npm/@tensorflow/tfjs@1.3.1/dist/tf.min.js"></script>
<script src="https://cdn.jsdelivr.net/npm/@tensorflow-models/speech-commands@0.4.0/dist/speech-commands.min.js"></script>

<script type="text/javascript">
  // more documentation available at
  // https://github.com/tensorflow/tfjs-models/tree/master/speech-commands

  // the link to your model provided by Teachable Machine export panel
```

Copy

⊕ Añadir una clase

Ruido de fondo ?

20 muestras de audio /

Micrófono Subir

hola

15 muestras de audio /

Micrófono Subir

adios

15 muestras de audio /

Micrófono Subir

Añadir una clase

Export your model to use it in projects.

Tensorflow.js TensorFlow Lite

Export your model:

☒ Upload (shareable link) ☐ Download

Your sharable link:

<https://teachablemachine.withgoogle.com/models/9cu5kfHH3/> Copy

When you upload your model, Teachable Machine hosts it at this link for free. (FAQ: [Who can use my model?](#))

✓ Your cloud model is up to date.

Code snippets to use your model:

JavaScript p5.js [Contribute on Github](#)

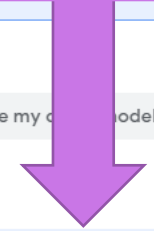
Learn more about how to use the code snippet on [github](#).

```
<div>Teachable Machine Audio Model</div>
<button type="button" onclick="init()">Start</button>
<div id="label-container"></div>
<script src="https://cdn.jsdelivr.net/npm/@tensorflow/tfjs@1.3.1/dist/tf.min.js"></script>
<script src="https://cdn.jsdelivr.net/npm/@tensorflow-models/speech-commands@0.4.0/dist/speech-commands.min.js"></script>

<script type="text/javascript">
  // more documentation available at
```

Copy

This is the URL of your trained model, which you will send through Canvas



YOU WILL ALSO HAVE TO UPLOAD THE CONFUSION MATRIX

The screenshot displays a machine learning interface with the following components:

- Class 1**: Labeled "Class 1" with an edit icon. It contains "3 Image Samples" and options for "Webcam" and "Upload". Two cat images are shown.
- Class 2**: Labeled "Class 2" with an edit icon. It features a "File" upload section with instructions: "Choose images from your files, or drag & drop here" and "Import images from Google Drive". It also shows "10 Image Samples" of various dogs.
- Training Panel**: Contains a "Train Model" button and an "Advanced" section with the following settings:
 - Epochs: 50
 - Batch Size: 16
 - Learning Rate: 0.001
 - Buttons for "Reset Defaults" and "Under the hood" (indicated by a purple arrow).
- Preview Panel**: Includes a "Preview" button, an "Export Model" button, and a message: "You must train a model on the left before you can preview it here."
- Footer**: Shows a language selector set to "English" and a version number "release-2-4-5".

Teachable Machine

Class 1

3 Image Samples



Webcam



Upload



Class 2

10 Image Samples



Webcam



Upload



+ Add a class

Training

Model Trained

Advanced

Epochs: 50

Batch Size: 16

Learning Rate:

0.001

Reset Defaults

Under the hood

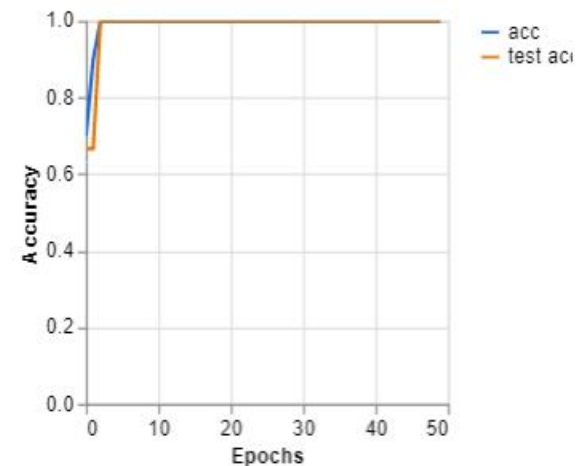
Accuracy per class

Calculate accuracy per class

Confusion Matrix

Calculate confusion matrix

Accuracy per epoch



Loss per epoch



YOU HAVE TO UPLOAD A SCREENSHOT OF THE MATRIX

Class 1

3 Image Samples

Webcam

Upload

Class 2

10 Image Samples

Webcam

Upload

Add a class

Training

Model Trained

Advanced

Epochs: 50

Batch Size: 16

Learning Rate: 0.001

Reset Defaults

Under the hood

Accuracy per class

CLASS	ACCURACY	# SAMPLES
Class 1	1.00	1
Class 2	1.00	2

Confusion Matrix

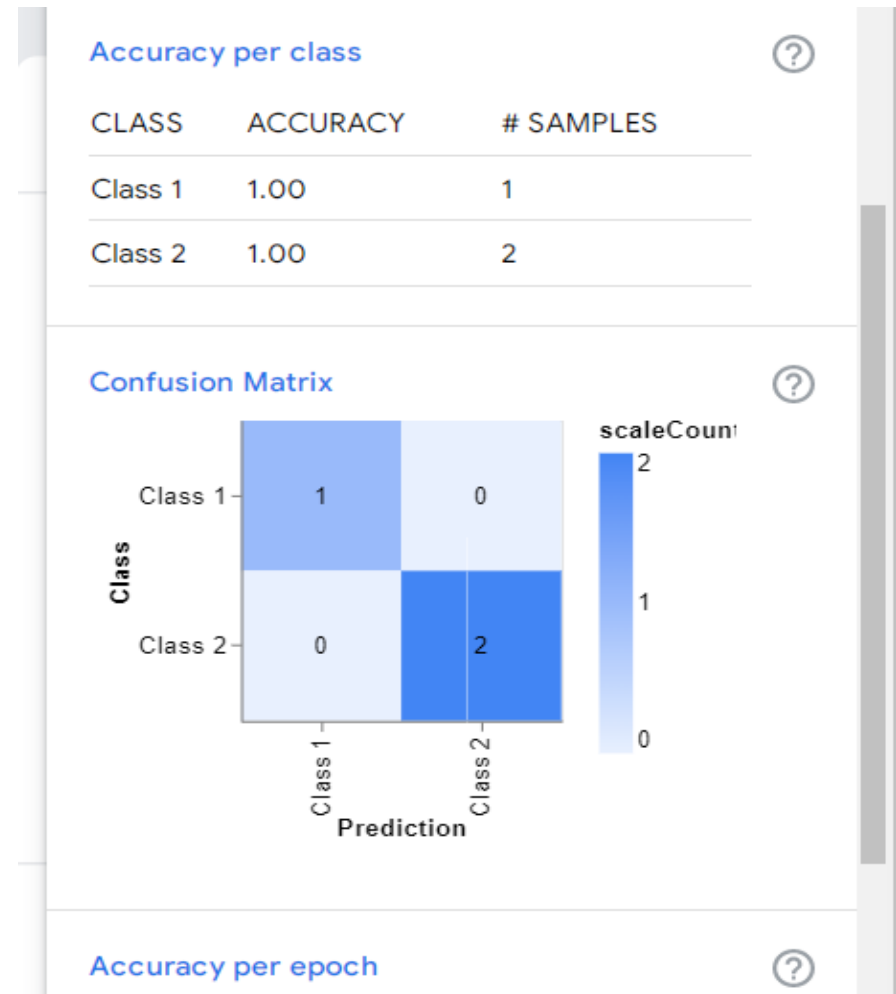
	Class 1	Class 2
Class 1	1	0
Class 2	0	2

scaleCount
2
1
0

Accuracy per epoch

— acc
— test acc

YOU HAVE TO UPLOAD A SCREENSHOT OF THE MATRIX



REAL-WORLD APPLICATION

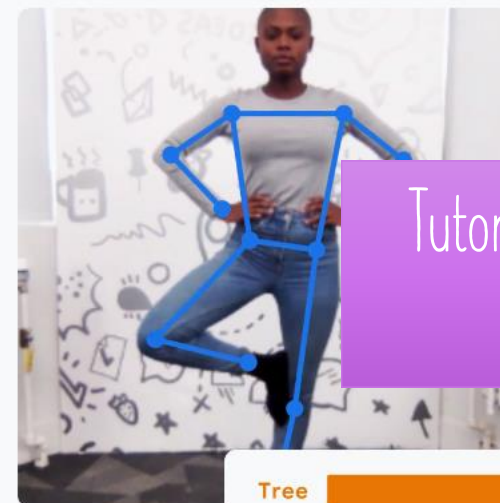
- Considering what can be done, think of a real-world application. Do not just focus on the classification, think what you would do with the results of the network if you could connect the result to another System, which would take as a starting point the result of the neural network.
- It doesn't have to be a complicated application! Think about some of the examples we've talked about in class.

 Descargar proyecto como archivo

Enviar comentarios

Una forma rápida y sencilla de crear modelos de aprendizaje automático para tus sitios web, aplicaciones y mucho más, sin necesidad de conocimientos especializados ni de programar.

Primeros pasos



Category	Percentage
Tree	100%
Wings	

Tutorials and Recommended Reading

Tutorials and Recommended Reading

¿Qué es Teachable Machine?

OTHER VIDEOS AND REFERENCE MATERIALS

1. <https://www.youtube.com/playlist?list=PLJfHZtseuscuTQfodmFnbZ3rBgCWsRT9t> These are very short videos from Google explaining how to collect data and train Teachable Machine

2. Videos showing Teachable Machine in Action and how to use the models as part of other projects (this last part is sadly too advanced for you now, but as soon as you learn to code you can do similar things! The presenter is a very enthusiastic person, and the videos are in English):

Part 1: Imágenes <https://www.youtube.com/watch?v=kwcillcW0g0>

Part 2: Poses <https://www.youtube.com/watch?v=UPgxnGC8oBU>

Part 3: Sonidos <https://www.youtube.com/watch?v=TOrVsLklltM>